# SST Non-Release Speculative Research Appendix v0.1

Composite Links, Phase Products, Modulo Classes, and Closed Modulation Ansätze

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**Non-release status.** This document records exploratory constructions, geometric intuitions, and unvalidated ansätze. Its contents are not part of the SST Canon, are not part of the release research-track companion, do not define SST predictions, and may be modified or rejected without requiring a Canon version change. No item in this appendix may be used to tune a Canon calibration, optimizer, null test, or stopping rule before formal promotion.

## 1 Epistemic ledger and promotion rule

| **Item** | Non-release status |
|---|---|
| **Periodic exponential modulation** | `[ANSATZ]`; one closed modulation family among many. |
| **Torus link** $T(6,9)$ | `[RESEARCH CANDIDATE]`; exact three-component topology, no established SST selection law. |
| **Three components times three phase labels** | `[STRUCTURAL ANSATZ]`; naturally $\mathbb{Z}_3 \times \mathbb{Z}_3$, not automatically $\mathbb{Z}_9$. |
| **Order-nine carry dynamics** | `[SPECULATIVE]`; requires an explicit physical coupling law. |
| **Doubling map modulo nine** | `[EXACT DISCRETE MATHEMATICS / PHYSICAL STATUS OPEN]`. |
| **Rodin-style toroidal projection** | `[INSPIRATION ONLY]`; geometry depends on representation choices. |

Promotion into the release research track requires, at minimum:

1. independently defined physical variables and an evolution law;
2. dimensional consistency and conservation-law compatibility;
3. comparison against alternative moduli, geometries, and phase laws;
4. a preregistered falsifier and a minimal analytic or numerical test;
5. no use of the candidate pattern in calibration or model selection;
6. reproducible improvement over simpler control models.

## 2 Periodic exponential modulation

A bounded periodic radius modulation may be written as

$$\rho(u) = \rho_0 \exp[\eta \sin(nu)], \qquad n \in \mathbb{Z}, \tag{1}$$

with a torus-knot or torus-link parameterization

$$\mathbf{X}_{p,q,n,\eta}(u) = \begin{pmatrix} [R + \rho(u)\cos(qu)]\cos(pu) \\ [R + \rho(u)\cos(qu)]\sin(pu) \\ \rho(u)\sin(qu) \end{pmatrix}, \qquad 0 \le u < 2\pi. \tag{2}$$

For integer $p, q, n$, the parameterization is closed. Closure alone does not establish preservation of knot type, finite reach, adequate core clearance, or dynamical stability. The parameter $\eta$ must remain free in the primary test. Assignments such as $\eta = \ln \phi$ are secondary hypotheses and are not permitted in an optimizer before the free-$\eta$ comparison.

The mandatory controls include an unmodulated embedding, ordinary sinusoidal or Fourier modulation, a spline family with comparable degrees of freedom, and equalized length or energy constraints. The exponential family is promoted only if it produces a reproducible dynamical advantage not explained by parameter count or geometric clearance.

## 3 The $T(6,9)$ three-trefoil candidate

For the torus link $T(p,q)$, the number of components is

$$d = \gcd(p,q). \tag{3}$$

Consequently,

$$\gcd(6,9) = 3, \tag{4}$$

and each component has reduced torus-knot type

$$T\left(\frac{6}{3}, \frac{9}{3}\right) = T(2,3), \tag{5}$$

so $T(6,9)$ consists of three trefoil components. With consistent orientation, the pairwise linking number of parallel torus-link components is

$$\mathrm{Lk}_{ij} = \frac{pq}{d^2} = 6. \tag{6}$$

These are topological facts. They do not imply that $T(6,9)$ is selected by Euler or Biot–Savart dynamics, that it is more stable than alternative three-component links, or that it realizes a particle state.

**Hypothesis 3.1** (Composite-link candidate)**.** *After a single resolved trefoil reaches at least the declared numerical KAM–3 gate under a fixed protocol, $T(6,9)$ may be compared with control three-component links under matched core radius, circulation, total length, helicity metadata, and minimum clearance.*

## 4 Three-by-three phase labels

Label the three components by $a \in \mathbb{Z}_3$ and three local phase sectors by $k \in \mathbb{Z}_3$. The nine labels form

$$(a,k) \in \mathbb{Z}_3^{\text{component}} \times \mathbb{Z}_3^{\text{phase}}, \qquad |\mathbb{Z}_3 \times \mathbb{Z}_3| = 9. \tag{7}$$

This product group is not cyclic:

$$\mathbb{Z}_3 \times \mathbb{Z}_3 \ncong \mathbb{Z}_9, \tag{8}$$

because every nonzero element of $\mathbb{Z}_3 \times \mathbb{Z}_3$ has order three, whereas $\mathbb{Z}_9$ contains elements of order nine.

A genuine order-nine orbit therefore requires an additional successor law. One possible bookkeeping map is the base-three carry operation

$$S(a,k) = \begin{cases} (a+1,k), & a = 0,1, \\ (0, k+1 \pmod 3), & a = 2, \end{cases} \qquad S^9 = \mathrm{id}\,. \tag{9}$$

Equation (9) is only a discrete indexing rule until a physical interaction derives it.

## 5 Doubling modulo nine

The map

$$x_{m+1} = 2x_m \pmod 9 \tag{10}$$

is an exact finite dynamical system. Its orbit decomposition is

$$1 \to 2 \to 4 \to 8 \to 7 \to 5 \to 1, \tag{11}$$

$$3 \leftrightarrow 6, \tag{12}$$

$$0 \to 0. \tag{13}$$

The six-cycle is the multiplicative unit group

$$\mathbb{Z}_9^\times = \{1,2,4,5,7,8\}, \qquad \mathrm{ord}_9(2) = 6. \tag{14}$$

This is rigorous number theory. The familiar decimal digital-root representation follows from $10 \equiv 1 \pmod 9$ and is therefore partly tied to the choice of base ten. A physical role for Eq. (10) requires an independently derived nine-state variable and a reason that one physical update acts as multiplication by two modulo nine.

## 6 Separation from the KAM release programme

The v0.8.25 KAM release programme tests relative equilibria, reduced Floquet ellipticity, Krein signatures, nonlinear twist, finite-order non-resonance, Poincaré maps, action drift, and invariant-torus survival. The constructions in this appendix may not be used to preselect a stable state. The valid order is

$$\text{unbiased KAM search} \longrightarrow \text{observed structure} \longrightarrow \text{post-hoc comparison with speculative candidates.} \tag{15}$$

A candidate is rejected when its apparent advantage disappears under matched controls, resolution refinement, alternative parameterizations, or removal of base-dependent encoding choices.

**Child-level analogy.** The appendix is a sketchbook, not the construction manual. A beautiful gear pattern can suggest what to test, but the machine must still work when the pattern is hidden from the engineer.

## References

[1] D. Rolfsen, 1976, *Knots and Links*, Publish or Perish, reprinted by AMS Chelsea Publishing, https://bookstore.ams.org/chel-346.

[2] G. H. Hardy and E. M. Wright, 2008, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, https://global.oup.com/academic/product/an-introduction-to-the-theory-of-numbers-9780199219865.

[3] H. K. Moffatt, 1969, "The Degree of Knottedness of Tangled Vortex Lines," *Journal of Fluid Mechanics*, vol. 35, no. 1, pp. 117–129, doi:10.1017/S0022112069000991.